

306. If $x * y = x^2 + y^2 - xy$, then the value of $9 * 11$ is

- (a) 93 (b) 103
(c) 113 (d) 121

307. If $a * b = \frac{ab}{a+b}$, find the value of $3 * (3 * -1)$.

- (a) -3 (b) -1.5
(c) -1 (d) $\frac{2}{3}$

308. If $a * b = 2a - 3b + ab$, then $3 * 5 + 5 * 3$ is equal to

- (a) 22 (b) 24
(c) 26 (d) 28

309. If $x \oplus y = x^2 + 2y$, what is the value of p if $4 \oplus (3 \oplus p) = 50$?

- (a) 4 (b) 7
(c) 8 (d) 12.5

310. If $a * b * c$ means $\frac{a+b}{c}$ for all numbers except 0,

then $(a * b * c) * a * b$ is equal to

- (a) 0 (b) 1
(c) $\frac{a+b+c}{ab}$ (d) $\frac{a+b+ac}{bc}$
(e) $\frac{ab+bc+ca}{a+b+c}$

Directions (Questions 311 to 313): Each question below consists of a number. Your task is to follow the following steps in order to find the result from the given alternatives. (M.C.A., 2005)

Step I: Multiply the number by 3 and add the square of the number.

Step II: Divide the result obtained after Step I by 5 and add 6.

Step III: Divide the result obtained after Step II by 3 and then square the resulting number.

311. 6 (a) 13 (b) 30

(c) 31.03 (d) 31.36

312. 2.5 (a) 8.10 (b) 8.51 (approx.)
(c) 18.05 (d) 80.50 (approx.)

313. 5 (a) 2.80 (b) 12.80
(c) 20.08 (approx.) (d) 21.80 (approx.)

314. If the operation $\wedge$ is defined by the equation $x \wedge y = 2x + y$, what is the value of a in $2 \wedge a = a \wedge 3$?

(Campus Recruitment, 2009)

- (a) 0 (b) -1
(c) 1 (d) 4

315. If $\oplus$ is an operation such that $a + b = \begin{cases} 2a, & \text{when } a > b \\ a + b, & \text{when } a < b \\ a^2, & \text{when } a = b \end{cases}$

then $\left[\frac{(5 \oplus 7) + (4 \oplus 4)}{3(5 \oplus 5) - (15 \oplus 11) - 3} \right]$ is equal to (C.P.O., 2007)

(a) $\frac{1}{3}$ (b) $\frac{2}{3}$

(c) $\frac{14}{13}$ (d) $\frac{14}{23}$

316. At Srinagar, starting at 9 a.m. on a certain day, snow began to fall at a rate of $1\frac{1}{4}$ inches every two hours until 3 p.m. If there was already $2\frac{1}{4}$ inches of snow on the ground at 9 a.m., how many inches of snow was on the ground at 3 p.m. that day?

(a) $3\frac{3}{4}$ (b) 6

(c) 7 (d) $7\frac{1}{2}$

317. A squirrel starts climbing up a tree at the speed of 6 metres a minute but after every 6 metres it slips down 4 metres in the next minute. It will be able to reach the top 120 metres high in (R.R.B., 2006)

(a) 20 minutes (b) 60 minutes

(c) 115 minutes (d) $1\frac{1}{2}$ hours

318. Ram had ₹ 1000 in his Savings Bank Account. Every month in the first week he needs money, so he withdraws ₹ 500, but by the end of the month, he deposits ₹ 750. After how many months the original amount will grow three times? (R.R.B., 2005)

(a) 6 months (b) 7 months

(c) 8 months (d) 9 months

319. The growth of a medicinal plant of height 1 metre, is 1 cm/day and it is growing straight vertically upward. A man cuts 6 cm from the top, at a regular interval of every 11 days. Its height will be 2 metres after (I.A.M., 2007)

(a) 220 days (b) 250 days

(c) 275 days (d) None of these

320. A stairway 10 ft. high is such that each step accounts for half a foot upward and one foot forward. What distance will an ant travel if it starts from ground level to reach the top of the stairway? (R.R.B., 2009)

(a) 30 ft. (b) 33 ft.

(c) 10 ft. (d) 29 ft.

321. The sum of all proper fractions whose denominators are less than or equal to 100, is

(a) $22\frac{1}{4}$ (b) 1925

(c) 2475 (d) None of these

322. The value of $\frac{1-x^4}{1+x} \div \frac{1+x^2}{x} \times \frac{1}{x(1-x)}$ is

(a) 1 (b) $1 - x^2$

(c) $\frac{1}{x}$ (d) $1 + x$

(e) None of these